

MS&E 242: Machine Learning for Algorithmic Trading

Final Project Instruction

Due: June 12th (Friday) by 11:00 PM.

Final Project Overview

The final project can be individual or completed in groups of up to three students. Your topic must have a clear focus on a financial problem and be connected to topics covered in class. Within those requirements, you may choose any methodology related to machine learning, statistical modeling, or optimization. The goal is to define a meaningful problem, analyze an appropriate dataset, apply one or more suitable models, evaluate the results carefully, and explain the findings from both numerical and financial/economic perspectives.

Suggested Report Components

The final report is expected to contain the following components:

- **Problem description and motivation.** Clearly state the question you study, the target variable or outcome of interest, and the financial/economic motivation.
- **Data description and preprocessing.** Describe the data source, sample period, data format, key variables, and main cleaning or transformation steps.
- **Exploratory data analysis.** Include summary statistics, tables, or figures to help readers understand the data and motivate your modeling choices.
- **Model description.** Briefly describe the machine learning, statistical, or econometric model(s) used. If applicable, describe the model architecture (briefly) or main specification.
- **Training, validation, and testing.** Explain how the data are divided for training, validation/model selection, and final testing. For time-series or financial data, avoid using future information when constructing the training set.
- **Parameter or hyperparameter selection.** State which parameters or hyperparameters are selected and how they are chosen, preferably using the validation set or cross-validation.
- **Performance evaluation.** Use at least three appropriate evaluation metrics. Report in-sample and out-of-sample performance when applicable, and compare your model with at least one reasonable benchmark.
- **Results and interpretation.** Explain the numerical results and discuss the financial, economic, or business insights. Positive and negative findings are both acceptable if they are explained carefully.
- **Limitations and conclusion.** Discuss the main limitations of your analysis and summarize the key methodology, results, and takeaways.

General Instructions

- The report should be clear, organized, and self-contained. A reader should be able to understand the problem, data, method, evaluation, and findings without reading your code.
- Page limit: minimum of 3 pages and maximum of 10 pages, excluding references and no appendices.
- Both the report and codes should be submitted to GradeScope.
- Figures and tables should have clear titles and should be discussed in the text.
- Please acknowledge all data sources, open-sourced codes, online materials, packages, and AI tools if they are used.
- A complicated model is not required. A strong project should have a clear question, appropriate data, reasonable methodology, careful evaluation, and meaningful interpretation. While benchmark comparison is important, outperforming the benchmark is not required for a successful project; thoughtful insights, careful analysis, and meaningful takeaways are equally important.